Task Management System – Requirements Analysis

1. Purpose / System Goal

The Task Management System helps individuals and teams to plan, organize, assign and track tasks efficiently.
It provides a simple way to manage daily work, improve productivity and meet deadlines.

2. Users / Stakeholders

- Individual Users – create and manage their own tasks
- Team Members – collaborate on tasks
- Admin (optional) – manage users and system settings

3. Key Benefits

- Organize work in a structured way
- Improve time management
- Increase team collaboration
- Track progress easily
- Reduce missed deadlines
- Save time and increase productivity

MVP

1. User Management

- Register
- Login / Logout
- Profile Update

Manage user accounts securely with registration, login and profile management.

2. Task Management

- Create Task
- View Task List
- Edit Task
- Delete Task

Create, update, delete and view tasks with details, title and description.

3. Task Status

- To Do
- In Progress
- Done

Track task progress using status: To Do, In Progress and Done.

4. Due Date

- Set Due Date
- View Upcoming Tasks

Set due dates for tasks and view upcoming deadlines.

5. Comments

- Add Comment on Task
- View Comments

Add comments to tasks for discussion and view all comments.

6. Notifications

- Due Date Reminder
- Task Assigned Notification
- Status Change Notification

Get notifications for due dates, task assignments and status changes.

4. Functional Requirements

- User can register a new account.
- User can login and logout.
- User can update their profile information.
- User can create a new task.
- User can view all their tasks.
- User can edit an existing task.
- User can delete a task.
- User can set status for a task (To Do, In Progress, Done).
- User can set a due date for a task.
- System can show upcoming tasks based on due date.
- User can add comments on a task.
- User can view all comments for a task.
- System can send due date reminder notifications.
- System can send task assigned notifications.
- System can send status change notifications.

Notes

This document describes the core (MVP) requirements of the Task Management System. Additional features can be added in future based on user needs.

Task Management System – Entity Thinking

1. What is Entity?

An entity is a real-world object or concept about which we store data in the system. Each entity has its own identity and contains related information.

2. How to Identify Entities?

- Read the requirements carefully
- Find the main objects (nouns)
- Identify what data we need to store
- Identify what data is related to each object
- Avoid storing actions or processes as entities

3. Rules for Good Entities

- Each entity must have a unique identity (Primary Key)
- Each entity represents one thing only
- No duplicate information in one entity
- Entities should be independent
- Easy to understand and maintain

4. Key Entities in Task Management System

User

Represents a person who can use the system. (Individual user or team member)

Example Data:
- Name
- Email
- Password

Task

Represents a piece of work that needs to be done.

Example Data:
- Title
- Description
- Status

Comment

Represents a comment or discussion on a task.

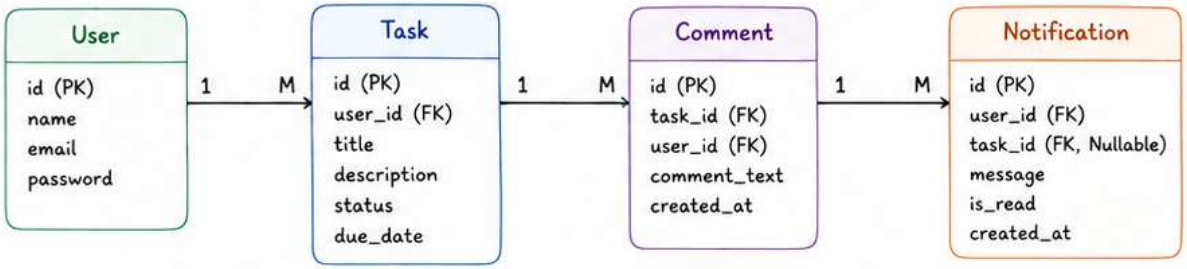
Example Data:
- Comment Text
- Created At

Notification

Represents a message sent to a user about a task or system event.

Example Data:
- Message
- Is Read
- Created At

5. Entity Relationship Overview



Relationship Explanation:

- One User can have Many Tasks, but one Task belongs to one User.
- One Task can have Many Comments, but one Comment belongs to one Task.
- One User can have Many Notifications, but one Notification belongs to one User.
- (Notification can be related to a Task, but it is optional.)

6. Important Points

- Focus on data (nouns), not on actions (verbs).
- Start with simple entities. Add more if needed later.
- Keep entities clear and independent.
- Every entity must have a Primary Key.
- Good entity design makes the database strong and flexible.

Relationship Diagram Notes

1. What is Relationship?

Relationship defines how entities (tables) are connected with each other in a database.

Example:

- User creates Tasks
- Task contains Comments

2. Why Relationships Important?

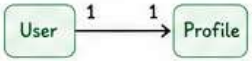
- Connects tables logically
- Reduces duplicate data
- Makes querying easier
- Maintains data consistency

3. Types of Relationships

1:1 One-to-One

One record connects to only one record.

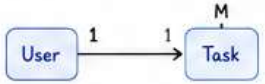
Example:
One User → One Profile



1:M One-to-Many

One record connects to multiple records.

Example:
One User → Many Tasks



Most common relationship.

M:N Many-to-Many

Multiple records connect to multiple records.

Example:
Many Users ↔ Many Teams

Requires a Junction Table.

Relationship Rules

Foreign Key Required

Relationships are created using Foreign Keys.

Example:
tasks.user_id → users.id

Parent → Child Thinking

Parent entity owns the child entity.

Example:
User → Task → Comment

Common Relationship Mistakes

- ✕ Missing Foreign Keys
- ✕ Wrong cardinality (1, M)
- ✕ Duplicate relationships
- ✕ Using Many-to-Many unnecessarily

5. Task Management System Relationships

User → Task

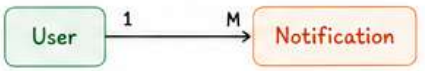
One User can create Many Tasks

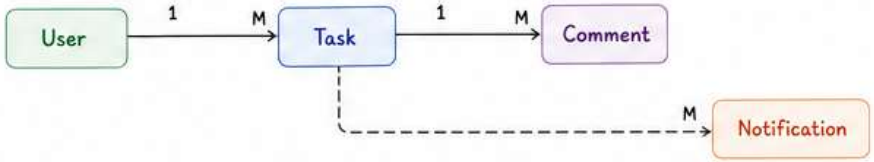
Task → Comment

One Task can have Many Comments

User → Notification

One User can receive Many Notifications



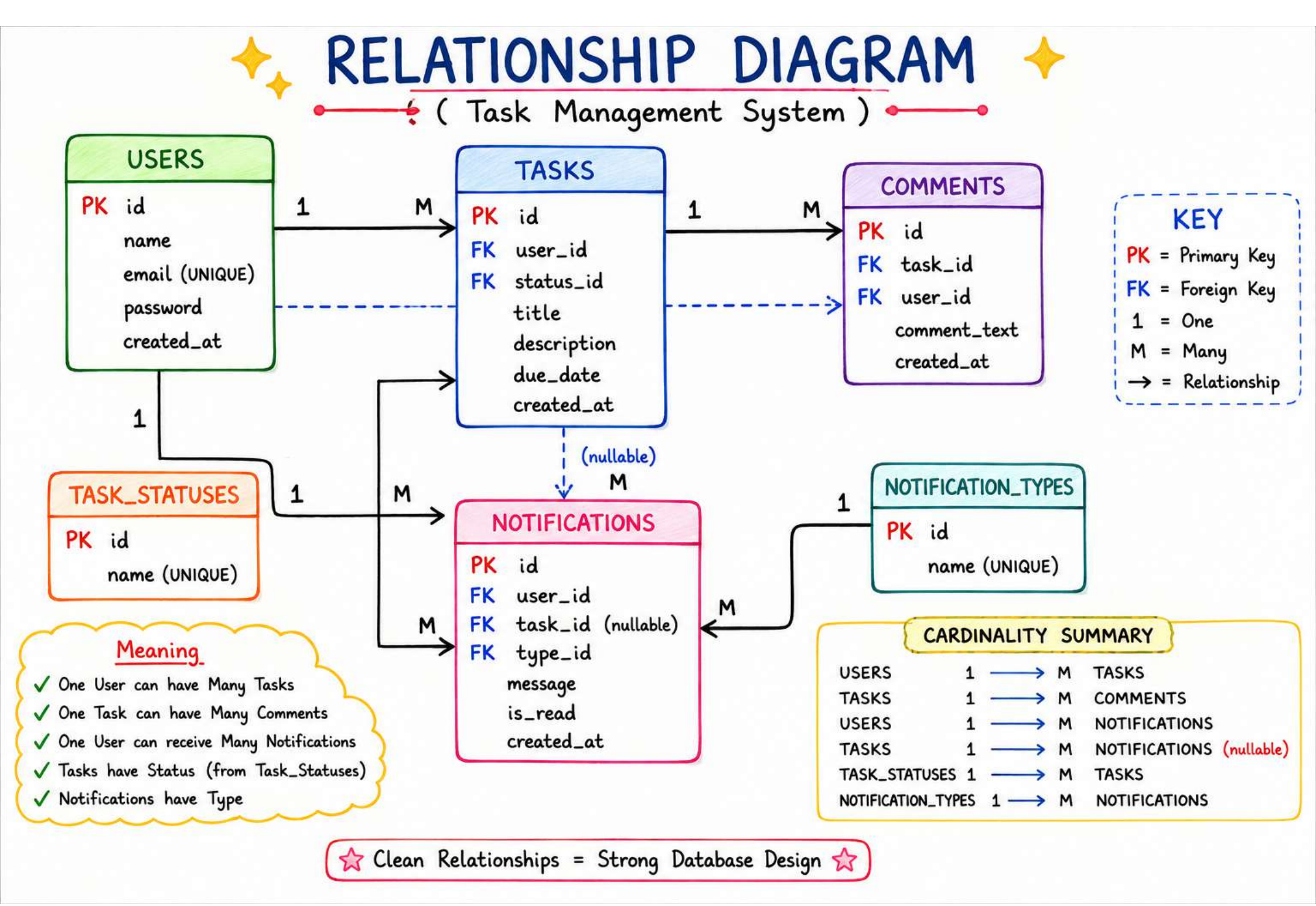


Symbols

1 = One
M = Many
→ = Relationship
FK = Foreign Key

7. Key Takeaways

- Relationships connect tables logically.
- Foreign Keys build relationships.
- One-to-Many is the most common.
- Many-to-Many requires a Junction Table.
- Good relationships improve database structure and performance.


RELATIONSHIP DIAGRAM
(Task Management System)
USERS
PK id
name
email (UNIQUE)
password
created_at
TASKS
PK id
FK user_id
FK status_id
title
description
due_date
created_at
COMMENTS
PK id
FK task_id
FK user_id
comment_text
created_at
KEY
PK = Primary Key
FK = Foreign Key
1 = One
M = Many
→ = Relationship
1
M
1
M
1
(nullable)
M
TASK_STATUSES
PK id
name (UNIQUE)
1
M
NOTIFICATIONS
PK id
FK user_id
FK task_id (nullable)
FK type_id
message
is_read
created_at
M
M
1
M
NOTIFICATION_TYPES
PK id
name (UNIQUE)
Meaning
✓ One User can have Many Tasks
✓ One Task can have Many Comments
✓ One User can receive Many Notifications
✓ Tasks have Status (from Task_Statuses)
✓ Notifications have Type
CARDINALITY SUMMARY
USERS 1 → M TASKS
TASKS 1 → M COMMENTS
USERS 1 → M NOTIFICATIONS
TASKS 1 → M NOTIFICATIONS (nullable)
TASK_STATUSES 1 → M TASKS
NOTIFICATION_TYPES 1 → M NOTIFICATIONS
Clean Relationships = Strong Database Design

ER Diagram HandNotes

1. What is ER Diagram?

ER Diagram (Entity Relationship Diagram) is a visual representation of the database structure.

It shows:

- Entities (Tables)
- Attributes (Columns)
- Relationships
- Primary Keys
- Foreign Keys

2. Purpose

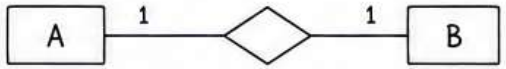
- Visualize database structure
- Understand relationships
- Easy planning & design
- Improve communication
- Reduce mistakes

3. Main Components

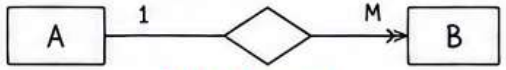
Component	Meaning	Symbol
Entity	Table / Object	▭
Attribute	Column / Field	⬭
Primary Key (PK)	Unique identifier	Attribute name
Foreign Key (FK)	Links to another table's PK	Attribute name
Relationship	Connection between entities	◇
Cardinality	1, M, or M:N	1 M M:N

4. Relationship Types

One-to-One (1:1)

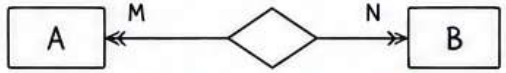


One-to-Many (1:M)



(Most Common)

Many-to-Many (M:N)



(Requires junction table)

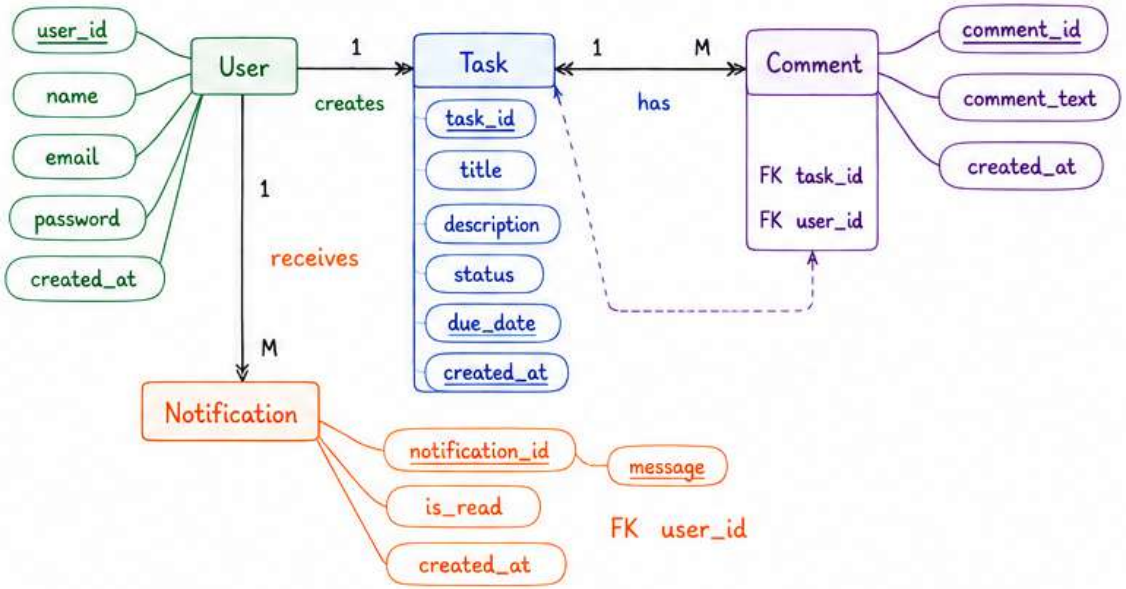
Task Management System – Entities & Attributes

User	Task	Comment	Notification
PK user_id	PK task_id	PK comment_id	PK ontification_id
name	FK user_id	FK task_id	FK user_id
email	title	FK user_id	message
password	description	comment_text	is_read
created_at	status	created_at	created_at
	due_date		
	created_at		

PK = Primary Key | Underlined = Primary Key

FK = Foreign Key | Dashed Underline = Foreign Key

6. Task Management System – ER Diagram



TASK MANAGEMENT SYSTEM – TABLE DESIGN

USERS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	User ID
name	VARCHAR(100)		NOT NULL	User Name
email	VARCHAR(150)		UNIQUE, NOT NULL	User Email
password	VARCHAR(255)		NOT NULL	Hashed Password
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

TASKS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Task ID
user_id	BIGINT	FK	NOT NULL	Reference to Users
status_id	BIGINT	FK	NOT NULL	Reference to Task Statuses
title	VARCHAR(255)		NOT NULL	Task Title
description	TEXT			Task Description
due_date	DATE			Due Date
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

TASK_STATUSES

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Status ID
name	VARCHAR(50)		UNIQUE, NOT NULL	Status Name (To Do, In Progress, Done)

COMMENTS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Comment ID
task_id	BIGINT	FK	NOT NULL	Reference to Tasks
user_id	BIGINT	FK	NOT NULL	Reference to Users
comment_text	TEXT		NOT NULL	Comment Text
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

NOTIFICATION_TYPES

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Type ID
name	VARCHAR(100)		UNIQUE, NOT NULL	Type Name (Due Reminder, Assigned etc.)

NOTIFICATIONS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Notification ID
user_id	BIGINT	FK	NOT NULL	Reference to Users
task_id	BIGINT	FK	NULL	Reference to Tasks (Nullable)
type_id	BIGINT	FK	NOT NULL	Reference to Notification Types
message	TEXT		NOT NULL	Notification Message
is_read	BOOLEAN		DEFAULT FALSE	Read Status
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

TASK MANAGEMENT SYSTEM – SQL SCHEMA TABLES

USERS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	User ID
name	VARCHAR(100)		NOT NULL	User Name
email	VARCHAR(150)		UNIQUE, NOT NULL	User Email
password	VARCHAR(255)		NOT NULL	Hashed Password
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

TASKS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Task ID
user_id	BIGINT	FK	NOT NULL	Reference to Users
status_id	BIGINT	FK	NOT NULL	Reference to Task Statuses
title	VARCHAR(255)		NOT NULL	Task Title
description	TEXT			Task Description
due_date	DATE			Due Date
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

TASK_STATUSES

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Status ID
name	VARCHAR(50)		UNIQUE, NOT NULL	Status Name (To Do, In Progress, Done)

COMMENTS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Comment ID
task_id	BIGINT	FK	NOT NULL	Reference to Tasks
user_id	BIGINT	FK	NOT NULL	Reference to Users
comment_text	TEXT		NOT NULL	Comment Text
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

NOTIFICATIONS

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Notification ID
user_id	BIGINT	FK	NOT NULL	Reference to Users
task_id	BIGINT	FK		Reference to Tasks (Nullable)
type_id	BIGINT	FK	NOT NULL	Reference to Notification Types
message	TEXT		NOT NULL	Notification Message
is_read	BOOLEAN		DEFAULT FALSE	Read Status
created_at	TIMESTAMP		DEFAULT CURRENT_TIMESTAMP	Created Time

NOTIFICATION_TYPES

Column	Type	Key	Constraints	Description
id	BIGINT	PK	NOT NULL	Type ID
name	VARCHAR(100)		UNIQUE, NOT NULL	Type Name (Due Reminder, Task Assigned, Status Changed, etc.)

RELATIONSHIP SUMMARY

- USERS 1 —< TASKS
- TASK_STATUSES 1 —< TASKS
- TASKS 1 —< COMMENTS
- USERS 1 —< COMMENTS
- USERS 1 —< NOTIFICATIONS
- TASKS 1 —< NOTIFICATIONS
- NOTIFICATION_TYPES 1 —< NOTIFICATIONS

TASK MANAGEMENT SYSTEM - SQL CREATE TABLE STATEMENTS

1. USERS TABLE

```sql
CREATE TABLE users (
  id BIGINT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

2. TASK_STATUSES TABLE

```sql
CREATE TABLE task_statuses (
  id BIGINT PRIMARY KEY,
  name VARCHAR(50) UNIQUE NOT NULL
);
```

3. TASKS TABLE

```sql
CREATE TABLE tasks (
  id BIGINT PRIMARY KEY,
  user_id BIGINT NOT NULL,
  status_id BIGINT NOT NULL,
  title VARCHAR(255) NOT NULL,
  description TEXT,
  due_date DATE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_tasks_user
      FOREIGN KEY (user_id) REFERENCES users(id),
  CONSTRAINT fk_tasks_status
      FOREIGN KEY (status_id) REFERENCES task_statuses(id)
);
```

4. COMMENTS TABLE

```sql
CREATE TABLE comments (
  id BIGINT PRIMARY KEY,
  task_id BIGINT NOT NULL,
  user_id BIGINT NOT NULL,
  comment_text TEXT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_comments_task
      FOREIGN KEY (task_id) REFERENCES tasks(id),

  CONSTRAINT fk_comments_user
      FOREIGN KEY (user_id) REFERENCES users(id)
);
```

5. NOTIFICATION_TYPES TABLE

```sql
CREATE TABLE notification_types (
  id BIGINT PRIMARY KEY,
  name VARCHAR(100) UNIQUE NOT NULL
);
```

6. NOTIFICATIONS TABLE

```sql
CREATE TABLE notifications (
  id BIGINT PRIMARY KEY,
  user_id BIGINT NOT NULL,
  task_id BIGINT NULL,
  type_id BIGINT NOT NULL,
  message TEXT NOT NULL,
  is_read BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

  CONSTRAINT fk_notifications_user
      FOREIGN KEY (user_id) REFERENCES users(id),

  CONSTRAINT fk_notifications_task
      FOREIGN KEY (task_id) REFERENCES tasks(id),

  CONSTRAINT fk_notifications_type
      FOREIGN KEY (type_id) REFERENCES notification_types(id)
);
```

NOTES

- All ID columns are BIGINT.
- Primary Key (PK) uniquely identifies each record.
- Foreign Key (FK) creates relationships between tables.
- created_at is used to track record creation time.
- is_read in notifications table tracks read/unread status.

LEGEND

CREATE TABLE	- SQL keyword
BIGINT	- Numeric data type
VARCHAR(n)	- Variable length string
TEXT	- Large text data
DATE	- Date value
TIMESTAMP	- Date and time
PRIMARY KEY	- Uniquely identifies a record
FOREIGN KEY	- Creates relationship
UNIQUE	- Ensures all values are different
NOT NULL	- Column cannot have NULL value
DEFAULT	- Sets default value if not provided
BOOLEAN	- True/False value

TASK MANAGEMENT SYSTEM – NORMALIZATION REPORT

GOAL : Eliminate redundant data and organize tables to ensure data integrity and consistency.

NORMAL FORM	RULE	CHECKING PROCESS	RESULT (OUR SCHEMA)	REMARKS
1NF	First Normal Form • Each column should have atomic (single) value. • No repeating groups / arrays.	✓ Checked all tables and columns. ✓ All values are atomic. ✓ No repeating groups or multi-valued attributes.	✓ All tables are in 1NF. All values are atomic and each record is unique.	No column contains multiple values. Each column holds single and indivisible value.
2NF	Second Normal Form • Must be in 1NF. • No partial dependency. Every non-key column must be fully dependent on the primary key.	✓ All tables have single column primary key (id). ✓ No table has composite key, so no partial dependency exists.	✓ All tables are in 2NF. Every non-key attribute is fully dependent on the primary key.	Since all tables use single column primary key, partial dependency issue does not exist.
3NF	Third Normal Form • Must be in 2NF. • No transitive dependency. Non-key column should not depend on another non-key column.	✓ Checked for transitive dependency in each table. ✓ All non-key attributes depend only on the primary key. Examples Checked : • users → no transitive dependency • tasks → no transitive dependency • comments → no transitive dependency • notifications → no transitive dependency • lookup tables → no transitive dependency	✓ All tables are in 3NF. No transitive dependency found in any table.	Lookup tables (task_statuses, notification_types) are used to avoid redundancy and transitive dependency.

NORMALIZATION IMPROVEMENTS

- Separated task statuses into task_statuses table.
- Separated notification types into notification_types table.
- This eliminates duplicate values.
- Ensures consistency and easier maintenance.

FINAL RESULT

All tables are normalized up to 3NF.
The database is well-structured, eliminates redundancy, and ensures data integrity.

✓ SCHEMA IS FULLY NORMALIZED

SUMMARY : Our Task Management System database is in 3NF.
It is efficient, consistent, and ready for development.